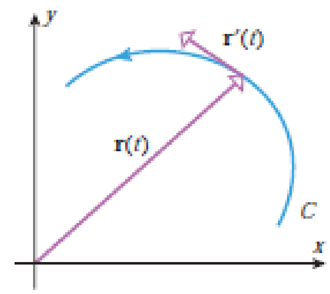
2.4 Unit Tangent, Normal, and Binormal Vectors

Tangent Vectors

Recall that in section 2.2 we discussed the idea of creating tangent lines to smooth vector-valued functions. Specifically, we stated the following definition.

Definition: Suppose that $\mathbf{r}'(t) \neq \mathbf{0}$. The vector $\mathbf{r}'(t)$ is called the tangent vector to the curve C at t. Therefore, we call, $\mathbf{T}(t) = \dfrac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|}$ the unit tangent vector.

Example 1: Find the unit tangent vector to the graph of $\mathbf{r}(t) = \cos(t)\mathbf{i} + \sin(t)\mathbf{j} + t\mathbf{k}$ at the point where $t = \dfrac{\pi}{4}$.



Normal Vectors

Recall that in section 2.2 we also introduced the following theorem.

Theorem: If $\mathbf{r}(t)$ is a differentiable vector-valued function in 2-space or 3-space and $\|\mathbf{r}(t)\|$ is constant for all t, then

$$\mathbf{r}(t) \cdot \mathbf{r}'(t) = 0$$

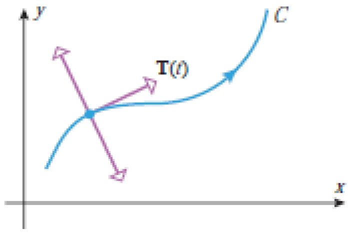
that is, $\mathbf{r}(t)$ and $\mathbf{r}'(t)$ are orthogonal vectors for all t.

Using this theorem we can conclude that since $\mathbf{T}(t)$ has a constant norm (that is, $\|\mathbf{T}(t)\|$ is constant) then $\mathbf{T}'(t)$ is orthogonal to it. This leads us to the following definitions.

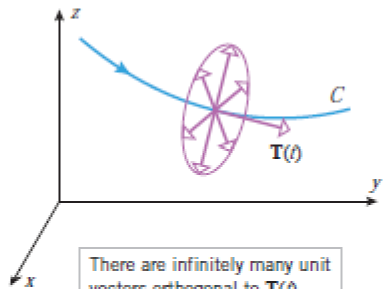
Definition: Provided that $\mathbf{T}'(t) \neq \mathbf{0}$, we call $\mathbf{T}'(t)$ a <u>normal vector</u> to the curve C at t.

Therefore, we call $\mathbf{N}(t) = \dfrac{\mathbf{T}'(t)}{\left\|\mathbf{T}'(t)\right\|}$ the principle unit normal vector to C at t.

Notice that there exist multiple normal vectors to a curve C and a given point t. Therefore, the idea of the principle unit normal vector is specifying a specific vector out of these several possibilities.



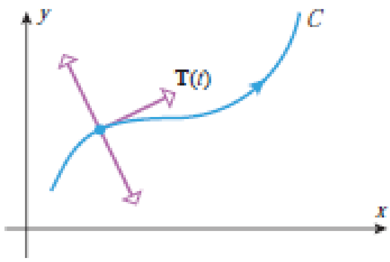
There are two unit vectors orthogonal to **T**(t).



There are infinitely many unit vectors orthogonal to **T**(t).

Note: As with the principle square root, if we are asked to find ***the*** unit normal vector, it is assumed to be the principle unit normal vector.

<u>**Example 2:**</u> Find the unit normal vector to $\mathbf{r}(t) = \cos(t)\mathbf{i} + \sin(t)\mathbf{j} + t\mathbf{k}$ at $t = \dfrac{\pi}{4}$.



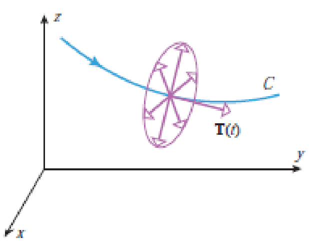
As we mentioned above there are actually many different normal vectors to a given curve. As shown above in 2-space there exists exactly 2 normal vectors to the curve *C* at the point *t*.

Theorem: Let *C* be a smooth curve in 2-space. The principle unit normal vector at each point along *C* is the unit normal vector that points *inward* (towards the concave side of the curve).

Proof: *(Shown in the notes on Blackboard)*

Note: In 2-space $\mathbf{N}(t)$ *is commonly referred to as the* *inward unit normal.*

Binormal Vectors



In 3-space there are infinitely many normal vectors at a point on *C*. $\mathbf{N}(t)$ is just one of these vectors, specifically the vector that when paired with $\mathbf{T}(t)$ determine a plane which is tangent to *C*. This plane is called the *osculating plane*.

Definition: If *C* is a graph of a vector-valued function $\mathbf{r}(t)$ in 3-space, then we define the binormal vector to *C* at *t* to be

Note: $\mathbf{B}(t)$ *is a unit vector.*

Example 3: Prove that $\mathbf{B}(t)$ is a unit vector.

Definition: A coordinate system determined by **T**, **N**, and **B** is called the **TNB**-frame or the Frenet-frame.

Theorem: The binormal vector $\mathbf{B}(t)$ can be expressed directly in terms of $\mathbf{r}(t)$ as:

$$\mathbf{B}(t)=\frac{\mathbf{r}'(t)\times\mathbf{r}''(t)}{\|\mathbf{r}'(t)\times\mathbf{r}''(t)\|}$$

Proof: Follows from the definitions of $\mathbf{B}(t)$, $\mathbf{N}(t)$, *and* $\mathbf{T}(t)$.

Example 4: Find the binormal vector to $\mathbf{r}(t)=\cos(t)\mathbf{i}+\sin(t)\mathbf{j}+t\mathbf{k}$ at $t=\frac{\pi}{4}$.